Advanced Statistics

Regression analysis - extensions of linear models

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Linear model extension

The univariate linear model can be extended in various ways

- Adding additionnal covariables
- Adding interactions between covariables
- Mathematical transformation of the covariables and/or the response

Multiple variable

We can extend the univariate linear model by adding additionnal covariables:

$$y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + ... + \epsilon$$

The interpretation of the regression coefficients slightly changes:

- β_0 is the expected y value when $X_1, X_2, ...$ are at 0
- β_1 is the expected unit change in y per unit change in X_1 while holding all other covariates at a fixed value.

Multiple variables

For instance say that we estimated the following model:

$$Height_i[m] = 20 + 2 * pH_i + 0.1 * Nitrogen[mg/g]$$

What do the regression coefficients (20, 2 and 0.1 mean)? Can we tell which variabe has the strongest effect from these values?

Multiple variables

Answer:

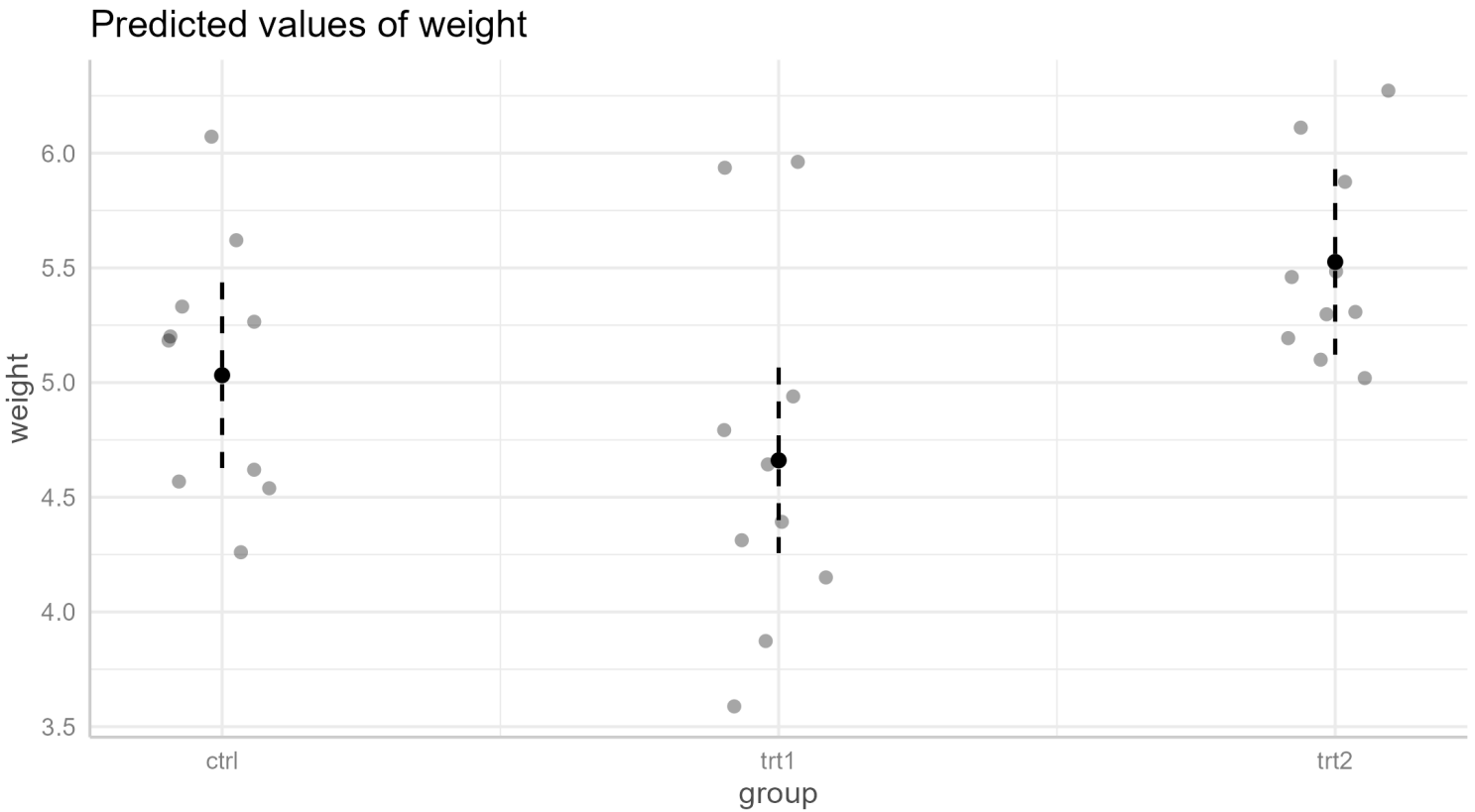
- Tree height is expected to be 20m for a pH of 0 and a nitrogen concentration of 0.
- Increasing the pH value by 1 increase tree height by 2m for a given nitrogen concentration.
- Increasing the nitrogen concentration by 1mg/g increase tree height by 0.1m for a given pH value.
- Given the different units of the covariables, we cannot directly compare the regression coefficients.
- To identify the (continuous) covariable with strongest effect irrespective of their units, we can scale them: $x^* = \frac{x-\mu}{\sigma}$, and use the scaled covariables in the regression model.

Multiple variable and collinearity

- Covariables have often some degrees of correlations between them
- This does not affect the estimated slopes (they remain unbiased) but the estimation become less precise (wider confidence intervals) which can be an issue in some circumstances
- Dropping collinear variables can be done, but the effect of the remaining covariables absorb the effects of the dropped ones
- For instance, if X_1 and X_2 have both an effect of 1 on the response y and a correlation of 0.7, only keeping X_1 in the model result in an estimated effect of around 1.7

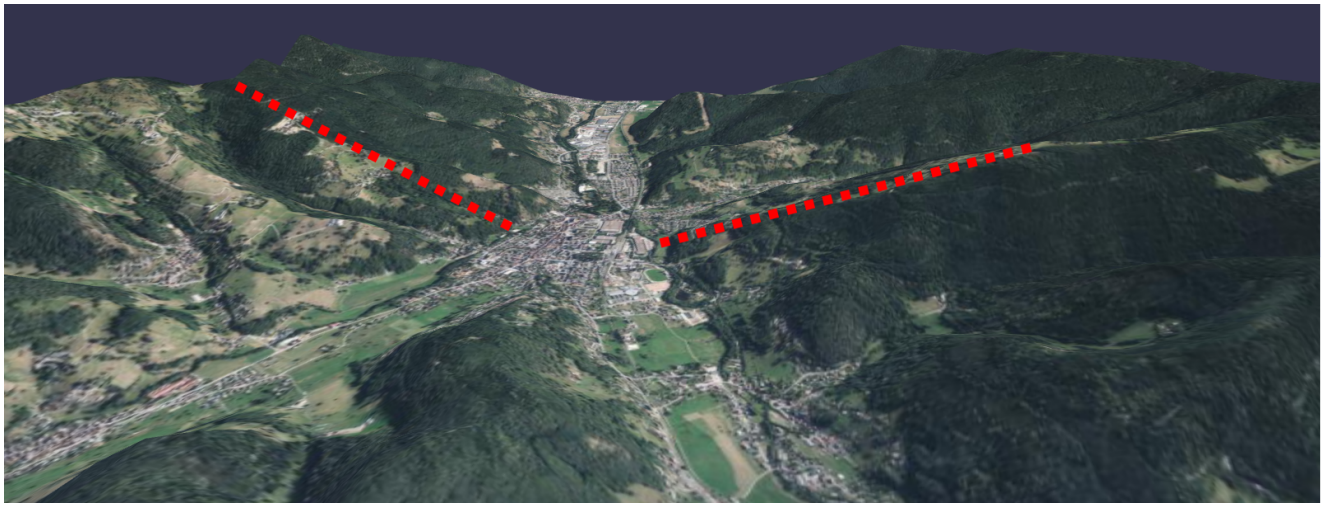
Apparté - categorical variables

Categorical covariables (like sex, species or treatment) can be added to regression models and are specified as contrasts - by default the treatment contrast is used, setting the first level as baseline.

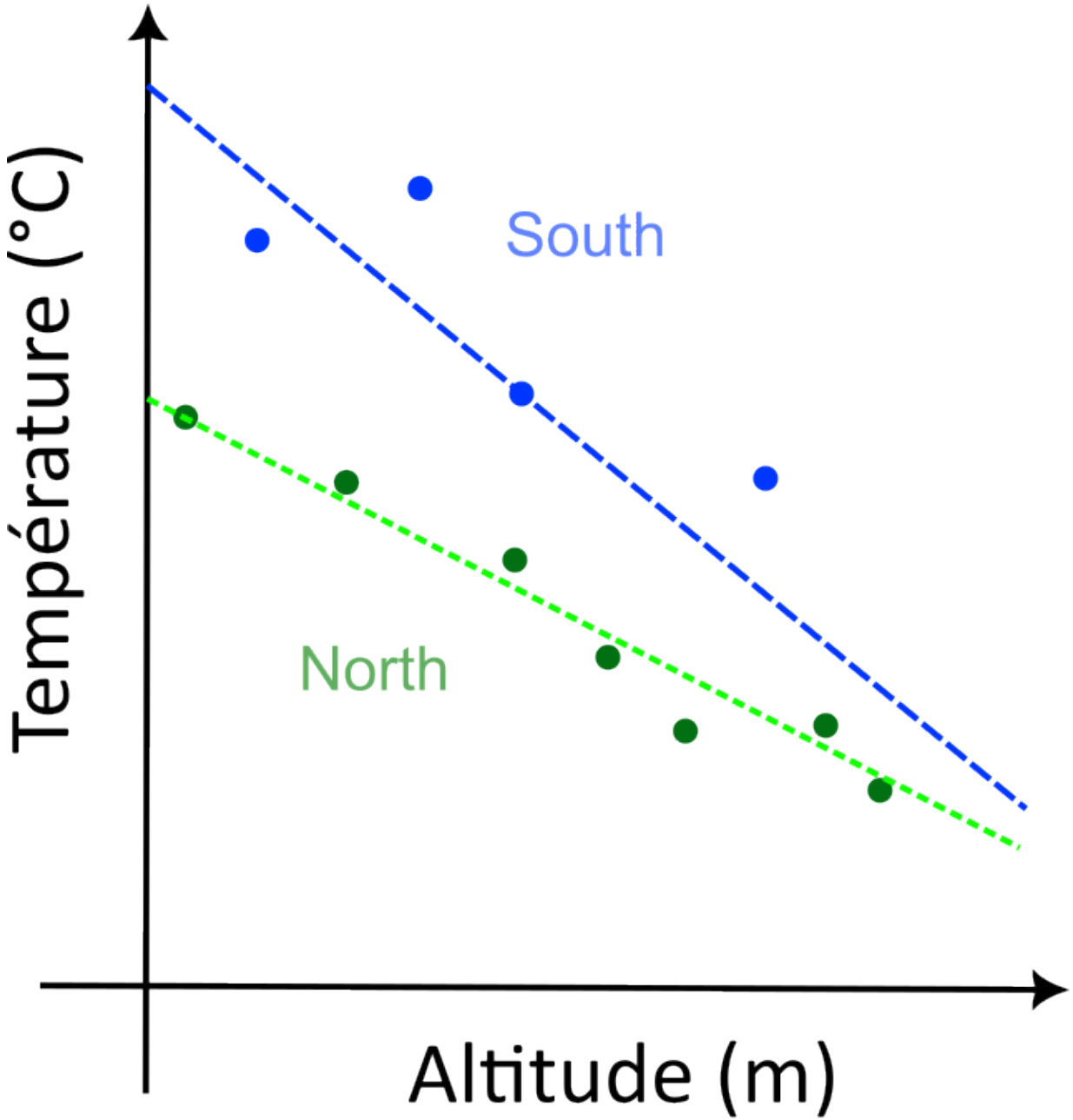


Interaction

With multiple covariables, the effect of one covariable depend on the value of another covariable, this is called an interaction.
For instance, the change in temperature along the altitudinal gradient can depend on the slope exposition (north vs south)



Interaction



Interaction

The corresponding linear model is the following:

$$\begin{aligned} temperature_i = \beta_0 + \beta_1 * I(exp_i == south) + \beta_2 * altitude_i \\ + \beta_3 * I(exp_i == north) * altitude_i + \epsilon \end{aligned} \tag{1}$$

The coefficient have the following interpretation:

- β_0: temperature at an altitude of 0m on the south-facing slope
- β_1: temperature difference between the north and south facing slope at an altitude of 0m
- β_2: temperature change with altitude on the south-facing slope
- β_3: difference in temperature change with altitude between the north and south facing slope

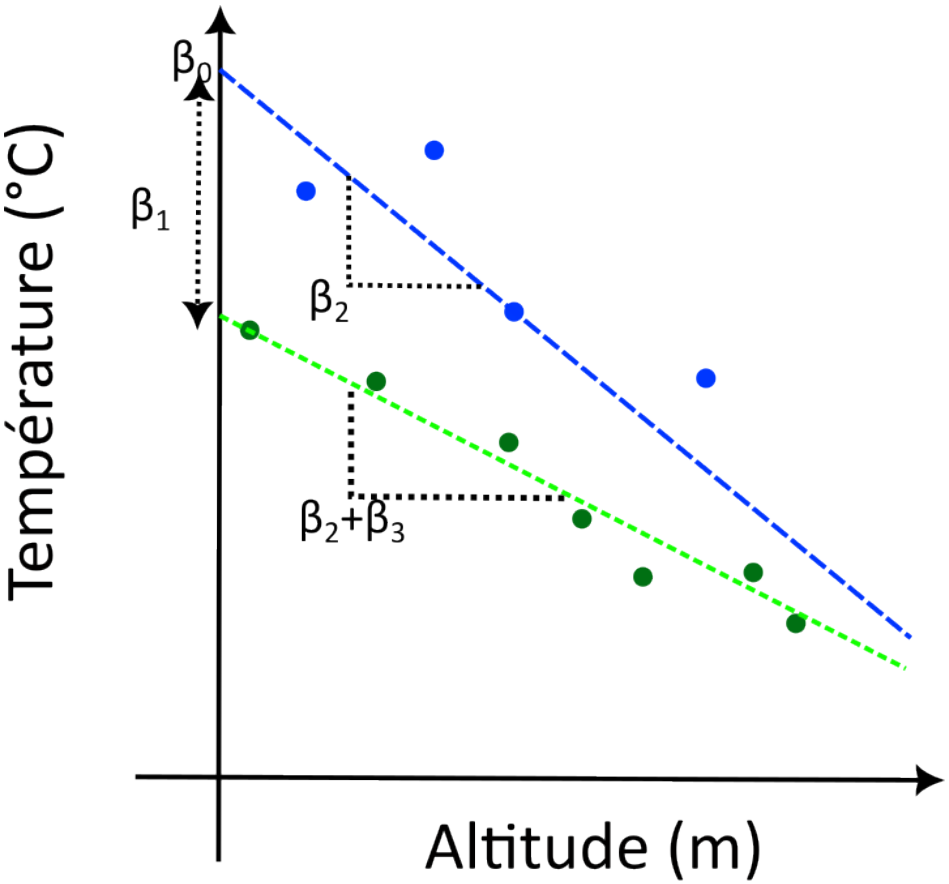
Interaction

We can also write this equation as follow:

$$temperature = \begin{cases} \beta_0 + \beta_2 * altitude & \text{if exp. is south} \\ (\beta_0 + \beta_1) + (\beta_2 + \beta_3) * altitude & \text{if exp. is north} \end{cases} \quad (2)$$

Interaction

And in a graph:



Question: what is the sign of β_1 and of β_3?

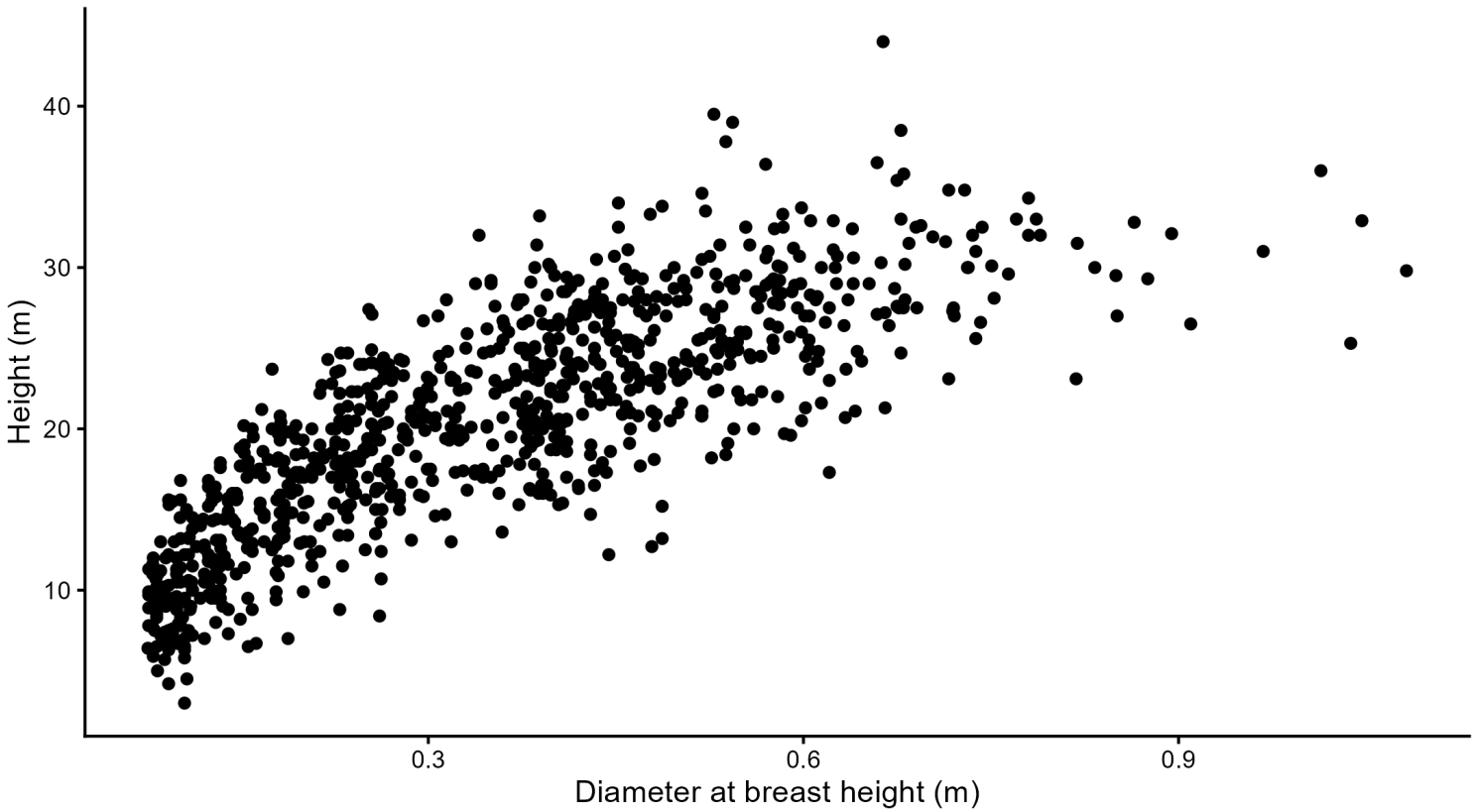
Interaction

- We saw the case of an interaction between a categorical and a continuous covariable
- We can also have continuous-continuous and categorical-categorical interactions
- We can also have interactions between three, four ... variables, the interpretation becomes harder as the interactions get more complex.

Going non-linear: transformation and polynomes

Many relations are non-linear, linear models allow to fit such relations through two main approaches:

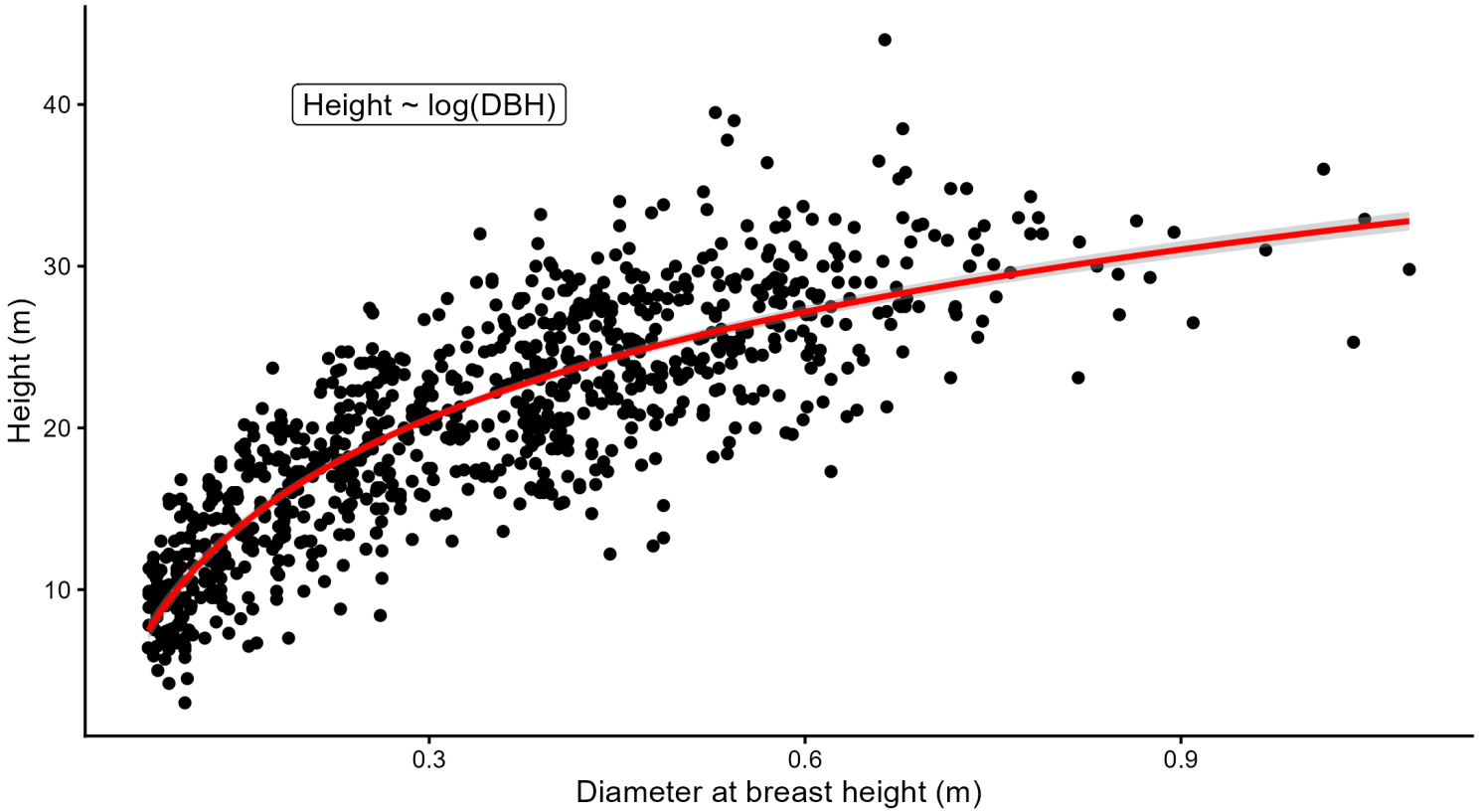
1. Variable transformation



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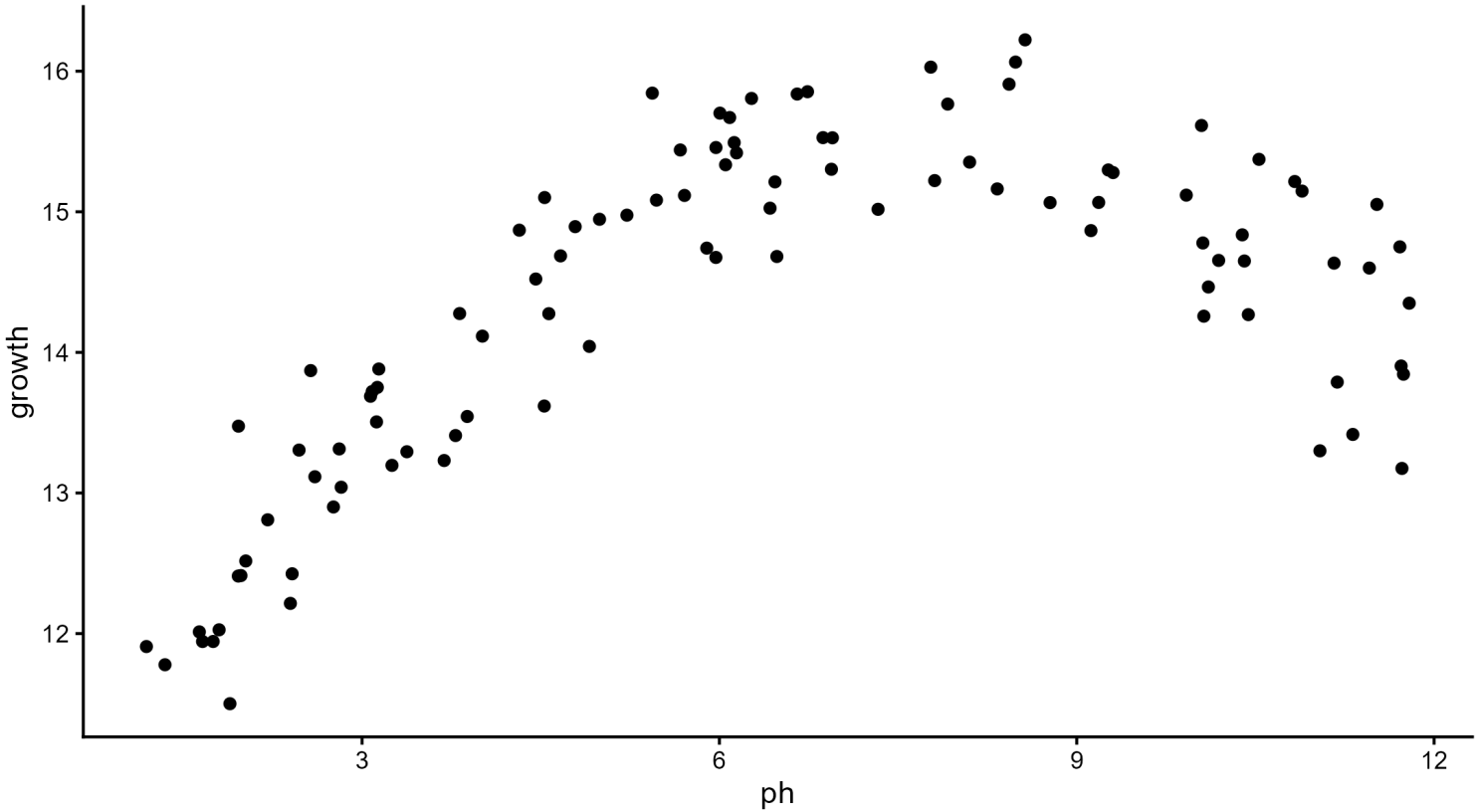
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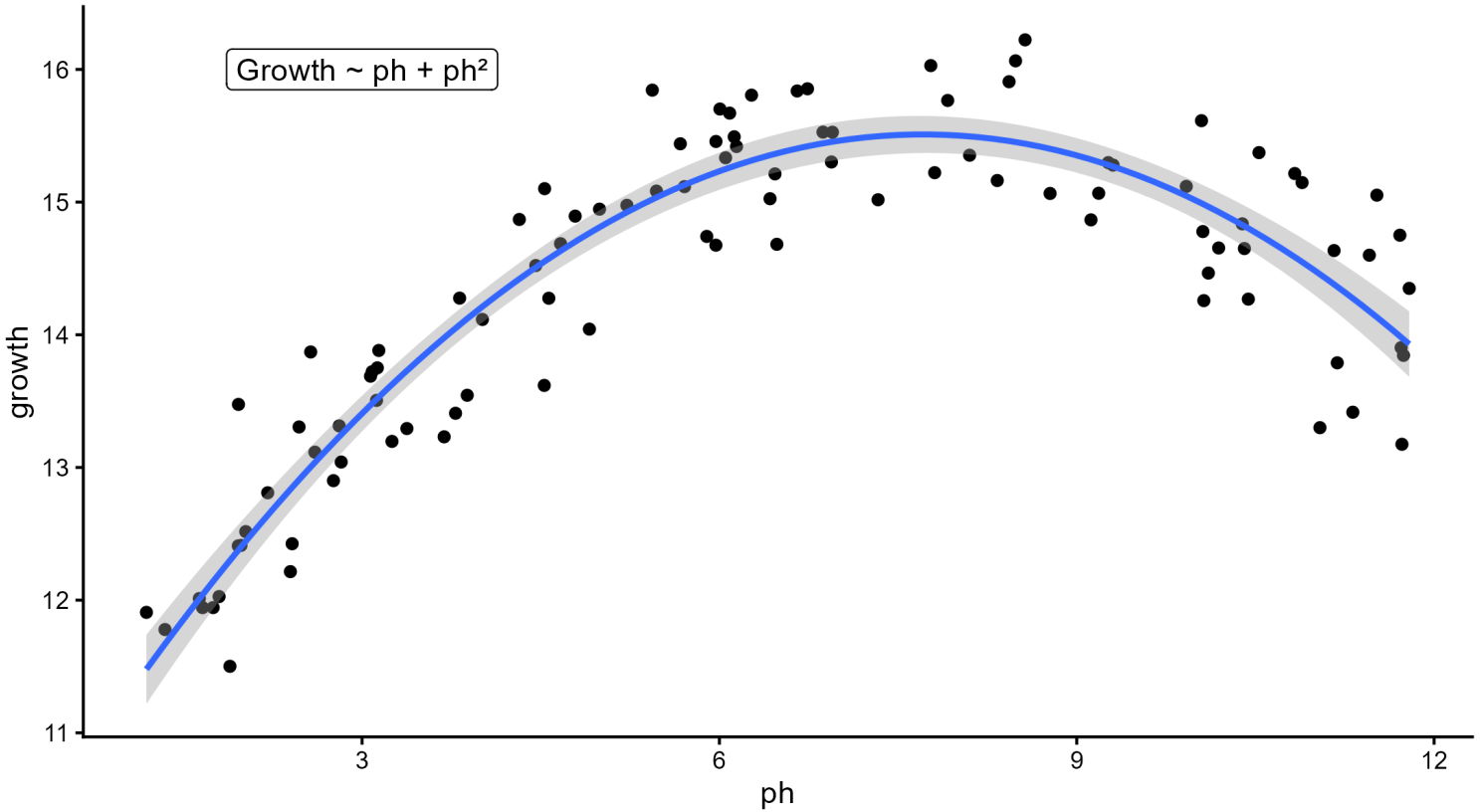
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2. Polynomes



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2. Polynomes



Coding time